

NIHARIKA BHASIN

133 Bayridge Ave, Brooklyn, NY 11220

📞 646-358-5226 ✉ nb4048@nyu.edu 🌐 [Portfolio](#) [linkedin.com/in/niharika-bhasin](#) [github.com/niharikabhasin](#)

Education

New York University

Masters in Computer Science

Sep 2024 – May 2026

Brooklyn, NY

SRM Institute of Science and Technology

Bachelors of Technology in Computer Science Engineering

Sep 2020 – May 2024

Delhi, India

Technical Skills

Languages: Python, C#, Java, C/C++, JavaScript, SQL, HTML/CSS

Frameworks: ASP.NET Core, .NET, FastAPI, Django, Node.js, React, Spring Boot

Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, Cognito, S3), Docker, Kubernetes, Terraform, CI/CD

Systems & Architecture: Distributed Systems, Microservices, Authentication, Event-Driven Architecture, REST APIs, System Design, Concurrency, Caching, gRPC, WebSockets

Data & Streaming: PostgreSQL, Redis, Kafka

Experience

Software Engineering Intern

Tassel

July 2026 – Present

New York City, NY

- Developing authentication infrastructure using **C#/.NET** and **Stytch**, implementing secure **user identity**, **onboarding**, **authorization**, and **session management** workflows across the platform.
- Extending and maintaining **distributed backend microservices**, contributing production features across service boundaries while improving **platform reliability**, scalability, and maintainability.
- Debugging and resolving **production backend issues** by tracing requests across microservices, identifying root causes, implementing fixes, and delivering changes through **peer-reviewed pull requests**.
- Designing and maintaining **unit** and **integration test suites** to validate backend functionality, improve code quality, and ensure reliable **production deployments**.

Software Engineering Intern

Toricent Labs

June 2025 – August 2025

New York City, NY

- Architected a scalable serverless backend on **AWS (Lambda, API Gateway, DynamoDB)** supporting **1M+ monthly API invocations**.
- Led a zero-downtime migration from Firebase to AWS using **Terraform**, reducing infrastructure costs by **30%**.
- Built **50+ production microservices** in Node.js and Python, integrating OAuth2 authentication while improving API performance through distributed caching.

SWE Graduate Assistant, NYU ECE

New York University

June 2025 – May 2026

Brooklyn, NY

- Built a centralized Lab Operations Portal using **FastAPI, React, and PostgreSQL** for 150+ researchers, automating equipment scheduling and inventory management.
- Optimized backend APIs, ETL pipelines, and RBAC security, reducing search latency by **45%** while streamlining administrative workflows by **35%**.

Projects

Safezone — Geospatial Safety Engine (Django REST, Docker, PostGIS, AWS)

- Architected a containerized full-stack system utilizing **PostgreSQL/PostGIS** for high-performance spatial indexing, reducing routing calculation overhead by **40%** through a **Mapbox-powered** engine secured with **AWS Cognito RBAC**.
- Engineered a reactive reporting workflow integrating **Cloudinary** for image processing and **Make.com webhooks** for asynchronous community alerts, validated via **Playwright E2E testing** to ensure **99.9%** deterministic API uptime.

CodeSync — Real-time Collaborative Editor (React, WebSockets, AWS SQS, Lambda)

- Built a low-latency code collaboration engine using **WebSockets** and **AWS SQS**, achieving **less than 50ms** sync delay across concurrent users.
- Deployed multi-tenant **Docker sandboxes** on isolated **EC2 nodes** for untrusted code execution, with automated **CI/CD via GitHub Actions**.

Ticket Finder — Distributed Microservices (Python, FastAPI, Redis, AWS)

- Designed a **high-concurrency distributed system** handling **10k+ concurrent requests** with **less than 100ms latency**, ensuring reliable ticket booking under heavy load.
- Implemented **dual-layer concurrency control** (**Redis locks + PostgreSQL row-level locking**) and **Saga pattern** for consistency across microservices.